

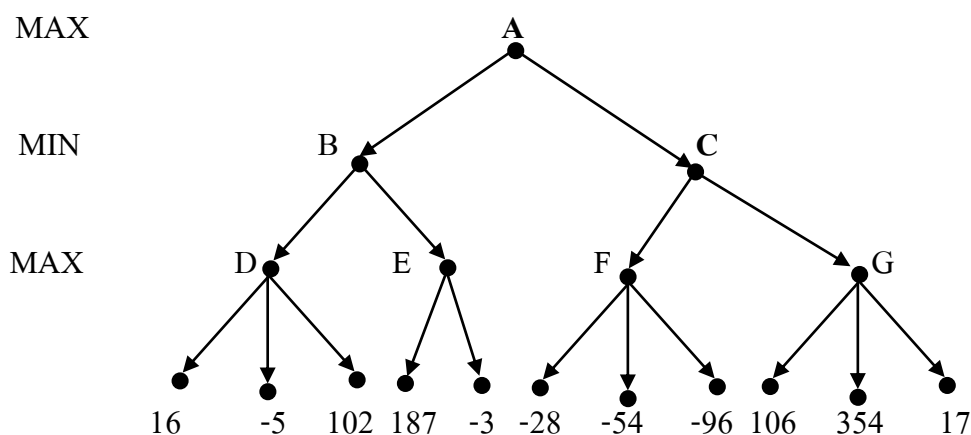


Faculty of Engineering
King Mongkut's Institute of Technology Ladkrabang
Final Examination, 1st Semester, Academic Year 2021
Subject: 13016223 Artificial Intelligence Year: 3
Examination Date: Monday 6th December 2021 Time: 13.30 – 16.30 Hrs.

Instructions: This is an OPENED BOOK exam

Questions:

1. (5 marks) Please explain why the iterative deepening search algorithm is complete and optimal.
2. (5+5 marks) Consider the game tree below where its root is a maximizing node, and the children are visited from left to right.
 - A. Compute the minimax value of nodes A, B, C, D, E, F and G using the minimax algorithm.
 - B. Demonstrate step by step how the tree is created and pruned by the alpha-beta algorithm.



3. (8+2 marks) A. Please draw a tree showing the sequence indicating how the A* algorithm finds an optimum answer for this 8-Puzzle where:

$h(n)$ is the sum of the (Manhattan) distances of the tiles from their goal positions, and $g(n)$ is the actual cost spending on moving from the Start node to node n .

the Start node is

2	3	
1	8	4
7	6	5

the Goal node is

1	2	3
8		4
7	6	5

B. Please also explain why this $h(n)$ is admissible.

4. (5 marks) Demonstrate using model checking whether the following sentence is true.

$$(A \rightarrow B) \wedge (B \rightarrow C) \models (A \rightarrow C)$$

5. (5 marks) Please construct a proof with reasons of each step of the proof, by applying some inference rules, to demonstrate that **V** can be derived from the following sentences

$$\begin{aligned} &\neg P \wedge \neg Q \\ &R \rightarrow Q \\ &(\neg R \wedge \neg P) \rightarrow (\neg S \wedge \neg W) \\ &S \leftrightarrow H \\ &H \vee V \vee W \end{aligned}$$

6. (5 marks) Given a knowledge base KB below, please demonstrate, using a proof by resolution, that $KB \models R$.

$$KB: (P \vee Q) \wedge (Q \rightarrow R) \wedge (\neg P \vee \neg S) \wedge (T \rightarrow S) \wedge (T)$$

7. (5 marks) Define a Prolog predicate **get_even(L1, L2)** where **L1** is a list of integers and **L2** is the list of all **even** integers extracted from **L1** in the similar sequence, for example:

```
?- get_even([1,2,3,4,5],L) .
L = [2,4] .
```

```
?- get_even([5,1],L) .
L = [] .
```

8. (5 marks) Please define the predicate **delete_all(X, L1, L2)** which deletes all elements **X** in list **L1** and returns list **L2** by **using ! (cut)**. Here is an example of a query evaluation,

```
?- delete_all(a, [a,b,a,c],X) .
X = [b,c];
no.
```

Information Page

$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$	commutativity of $\wedge$
$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$	commutativity of $\vee$
$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$	associativity of $\wedge$
$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma))$	associativity of $\vee$
$\neg(\neg\alpha) \equiv \alpha$	double-negation elimination
$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$	contraposition
$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$	implication elimination
$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$	biconditional elimination
$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$	De Morgan
$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$	De Morgan
$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$	distributivity of $\wedge$ over $\vee$
$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$	distributivity of $\vee$ over $\wedge$

Logical Equivalence

<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>
$\frac{p \quad p \rightarrow q}{\therefore q}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\frac{\neg q \quad p \rightarrow q}{\therefore \neg p}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\frac{p \vee q \quad \neg p}{\therefore q}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\frac{p}{\therefore p \vee q}$	$p \rightarrow (p \vee q)$	Addition
$\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Simplification
$\frac{p \quad q}{\therefore p \wedge q}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Rules of Inference